

Mawrooth (Saudi Identity Track)

An AI application that explains Saudi Nabati poetry and transforms it into visual representations

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Project Idea 01

Mawrooth is an AI application designed to make Saudi Nabati poetry more accessible to younger generations by explaining verses in clear language and transforming their poetic imagery into visual representations. The project adapts ALLaM-7B using QLoRA on a custom Nabati poetry dataset, supported by a specialized dictionary containing more than 1,300 Nabati terms to improve understanding of heritage vocabulary. Mawrooth provides an interactive experience that connects textual understanding with visual generation, offering a modern digital approach to exploring and preserving Saudi literary heritage.

Beneficiary Sectors 02

Mawrooth serves students, researchers, and heritage enthusiasts seeking a clearer understanding of Saudi Nabati poetry. It can also support cultural and educational institutions working to digitally document linguistic heritage and provide an interactive way to present Nabati poetry and its vocabulary to broader audiences, including those interested in Saudi culture and cultural tourism.

The Project's Role in Supporting and Advancing Arabic-Language AI Technologies 03

Mawrooth addresses a core challenge in Arabic AI: the dominance of Modern Standard Arabic and contemporary text in training data, which limits model performance on dialects and heritage vocabulary. The project introduces a scalable methodology built on three components: human-reviewed domain data, a flexible knowledge base that can be continuously expanded, and adaptation of a leading Arabic model (ALLaM-7B) via QLoRA instead of training from scratch. This methodology is not limited to Nabati poetry – it forms a replicable framework applicable to other Arabic dialects (Gulf, Egyptian, Maghrebi) and similar heritage domains, contributing to Arabic AI's ability to serve linguistic and cultural diversity.

The Role of AI and the Results 04

AI was applied by adapting ALLaM-7B using QLoRA on a custom Nabati poetry dataset, with a specialized dictionary integrated to support heritage vocabulary understanding. The model generates structured explanations in Modern Standard Arabic, which are then used to construct a visual description and generate an AI-created representation of the poetic scene. Evaluation on a held-out set of 59 examples achieved a token-level accuracy of 85.75%, with no evidence of overfitting.

Proposed Improvements and Future Development 05

Future development includes expanding the dataset to cover more diverse Saudi regions and poetic styles, improving model performance on unseen verses, and strengthening the connection between textual explanation and visual generation. Mawrooth could also be developed into a complete mobile application with educational features that help students and heritage enthusiasts explore Nabati poetry and its vocabulary.